AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2011

Computer Science Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL EIGHT questions. Questions 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

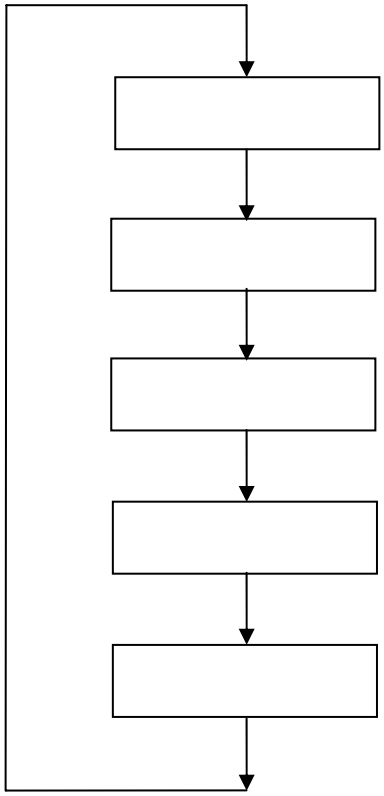
4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. The different stages of a software development cycle are given below. (2 Marks)

Implementation	Coding	Testing	Problem identification	Planning

Insert the given words to complete the diagram, showing the correct order.



b. Explain any THREE stages of software development that are given in Part (a). (3 Marks)

Q.2. (Total 5 Marks)

An airport has a number of hotels nearby. A data file is to be created to give customers information to allow them to select a hotel.

Write an appropriate variable that is used in BASIC language across each datum given in the table below.

Field	Data	Variable
Hotel reference	H41	
Name of the hotel	The Grand	
Number of rooms	45	
Hotel parking	Y	
Phone number	0061-47545556	

Q.3. (Total 5 Marks)

a. Consider the given program: (3 Marks)

```
10  CLS
20  TN=5
30  FOR COUNTER=1 TO 2
40
50  NEXT COUNTER
```

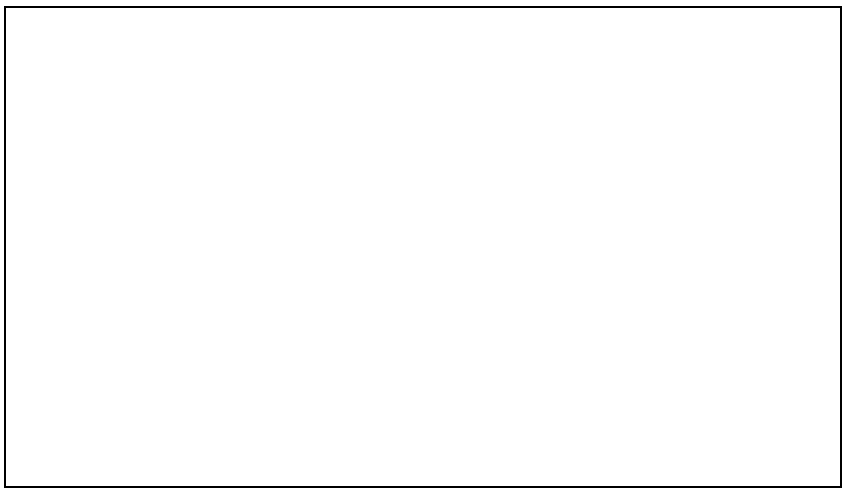
What will be the output if the given statements are inserted at line number 40?

I. PRINT COUNTER;
II. PRINT TN, COUNTER
III. PRINT TN; “x”; COUNTER;“=”; TN*COUNTER

I	II	III

b. What will be the output if the given statements are inserted at line number 20 and 40 of the program in part a? (2 Marks)

```
20  INPUT TN
40  PRINT TN; "x";COUNTER; "=";TN*COUNTER
```



Q.4. (Total 4 Marks)

Write a program to read marks of any four subjects using a single dimensional array which should calculate the total of entered numbers to produce marks sheet in ascending order.

Your program should produce the following output:

MARKS IN SUBJECT # 1 = 55
MARKS IN SUBJECT # 2 = 77
MARKS IN SUBJECT # 3 = 83
MARKS IN SUBJECT # 4 = 92
TOTAL **307**

Q.5. (Total 3 Marks)

Define the function for finding the result of the following equation:

X = X*X+3*X

The program should display the output using the value of X range from -3 to 3.

Q.6. (Total 3 Marks)

Complete the given truth table for the following Boolean expression:

$X = \overline{A}\ B\ \overline{C} + A\ \overline{B}\ C$

ROW	A	B	C	X
1	0	0	1	
2	0	1	0	
3	1	1	1	

Q.7. (Total 7 Marks)

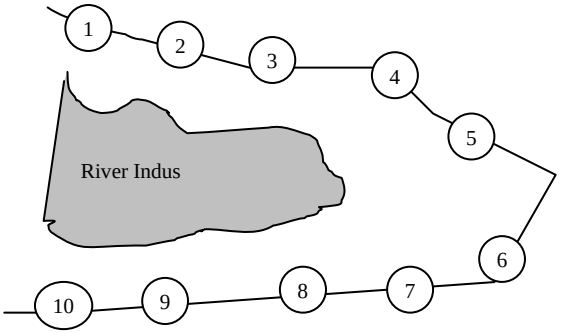
EITHER

a. A company has 5000 CDs, DVDs, videos and books in stock. Each item has a unique 5-digit code with the first digit identifying the type of item, i.e.
1 = CD
2 = DVD
3 = video
4 = book
In the code 15642, 1 identifies the item as a CD, and in the code 30055, 3 identifies the item as a video.

Write a program that:
- Inputs the codes for all 5000 items;
- Validates the code entered;
- Calculates how many CDs, DVDs, videos and books are in stock;
- Outputs the four totals.

OR

b. The following diagram shows a rail network.



The rail network consists of 10 stations. The fare between each station is Rs. 20. There is a 10% discount when 3 or more passengers travel together. Tickets can be purchased at any station using automated terminals.

Write a program for the automated terminals to:

- Input the starting station number, the destination station number and the number of passengers;
- Calculate the total fare and output the amount to be paid;
- Calculate the change (if any);
- Issue the rail ticket(s) and change.

Q.8. (Total 8 Marks)

EITHER

a. Discuss how viruses can be harmful for our computers and what can be done to keep the system protected.

OR

b. Discuss how robots have made changes in the manufacturing process of vehicles. What are some advantages and disadvantages of robots?

END OF PAPER

Please use this page for rough work

Please use this page for rough work